National Testing Agency

Question Paper Name :	B Tech 4th Apr 2026 Shift 1
Subject Name :	B. Tech
Creation Date :	2026-04-04 12:31:20
Duration :	180
Total Marks :	300
Display Marks:	Yes

B. Tech

Group Number :	1
Group Id :	6952784
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Id :	69527819
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527819
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 695278226 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x)=\cos^{-1}\left(\frac{4x+2[x]}{3}\right)$ is $[\alpha, \beta]$, then $12(\alpha+\beta)$ is equal to:

Options :

695278766. 6

695278767. 8

695278768. 9

695278769. 4

Question Number : 2 Question Id : 695278227 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the set of all solutions of $|x^2 + x - 9| = |x| + |x^2 - 9|$ is $[\alpha, \beta] \cup [\gamma, \infty)$, then $(\alpha^2 + \beta^2 + \gamma^2)$ is equal to:

Options :

695278770. 9

695278771. 18

695278772. 36

695278773. 72

Question Number : 3 Question Id : 695278228 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let z be a complex number such that $|z + 2| = |z - 2|$ and $\arg\left(\frac{z+3}{z-i}\right) = \frac{\pi}{4}$. Then

$|z|^2$ is equal to:

Options :

695278774. 9

695278775. 4

695278776. 5

695278777. 1

Question Number : 4 Question Id : 695278229 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of functions $f: \{1, 2, 3, 4\} \to \{a, b, c\}$, which are not onto, is:

Options :

695278778. 48

695278779. 45

695278780. 51

695278781. 35

Question Number : 5 Question Id : 695278230 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $S = \left\{ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \{0, 1, 2, 3, 4\} \text{ and } A^2 - 4A + 3I = 0 \right\}$

be a set of 2×2 matrices. Then the number of matrices in S, for which the sum of the diagonal elements is equal to 4, is:

Options :

695278782. 20

695278783. 17

695278784. 21

695278785. 19

Question Number : 6 Question Id : 695278231 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A=\begin{bmatrix}1 & 1 & 2\\ -2 & 0 & 1\\ 1 & 3 & 5\end{bmatrix}$. Then the sum of all elements of the matrix

$adj(adj\ (2(adjA)^{-1}))$ is equal to:

Options :

695278786. 3

695278787. 4

695278788. -4

695278789. -3

Question Number : 7 Question Id : 695278232 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The first term of an A.P. of 30 non-negative terms is $\frac{10}{3}$. If the sum of this A.P. is the cube of its last term, then its common difference is:

Options :

695278790. $\frac{5}{87}$

695278791. $\frac{25}{83}$

695278792. $\frac{15}{29}$

695278793. $\frac{5}{29}$

Question Number : 8 Question Id : 695278233 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of ways, of forming a queue of 4 boys and 3 girls such that all the girls are not together, is:

Options :

695278794. 5040

695278795. 3050

695278796. 3410

695278797. 4320

Question Number : 9 Question Id : 695278234 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3+(1+x)^4+(1+x)^5+...+(1+x)^{99}+(1+kx)^{100}$, $x \neq 0$, is $\left(43n+\frac{101}{4}\right)\left({}^{100}C_3\right)$ for some $n \in \mathbb{N}$, be p. Then the value of $p+n$ is:

Options :

695278798. 10

695278799. 11

695278800. 12

695278801. 13

Question Number : 10 Question Id : 695278235 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Suppose that the mean and median of the non-negative numbers 21, 8, 17, a, 51, 103, b, 13, 67, $(a > b)$, are 40 and 21, respectively. If the mean deviation about the median is 26, then $2a$ is equal to:

Options :

695278802. 109

695278803. 117

695278804. 161

695278805. 131

Question Number : 11 Question Id : 695278236 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the line $L_1 : x + 3 = 0$ intersect the lines $L_2 : x - y = 0$ and $L_3 : 3x + y = 0$ at the points A and B, respectively. Let the bisector of the obtuse angle between the lines L_2 and L_3 intersect the line L_1 at the point C. Then $BC^2 : AC^2$ is equal to:

Options :

695278806. 5:1

695278807. 1:5

695278808. 2:3

695278809. 3:2

Question Number : 12 Question Id : 695278237 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the vertex A of a triangle ABC be (1, 2), and the mid-point of the side AB be (5, −1). If the centroid of this triangle is (3, 4) and its circumcenter is (α, β), then $21(\alpha+\beta)$ is equal to:

Options :

695278810. 309

695278811. 403

695278812. 497

695278813. 524

Question Number : 13 Question Id : 695278238 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Suppose that two chords, drawn from the point (1, 2) on the circle $x^2 + y^2 + x - 3y = 0$ are bisected by the y-axis. If the other ends of these chords are R and S, and the mid point of the line segment RS is (α, β), then $6(\alpha + \beta)$ is equal to:

Options :

695278814. 1

695278815. 3

695278816. 4

695278817. 6

Question Number : 14 Question Id : 695278239 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A line with direction ratios 1, −1, 2 intersects the lines $\frac{x}{2} = \frac{y}{3} = \frac{z+1}{3}$ and $\frac{x+1}{-1} = \frac{y-2}{1} = \frac{z}{4}$ at the points P and Q, respectively. If the length of the line segment PQ is α, then $225\alpha^2$ is equal to:

Options :

695278818. 1024

695278819. 1014

695278820. 1104

695278821. 1204

Question Number : 15 Question Id : 695278240 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The square of the distance of the point (−2, −8, 6) from the line $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z}{-1}$ along the line $\frac{x+5}{1} = \frac{y+5}{-1} = \frac{z}{2}$ is equal to:

Options :

695278822. 3

695278823. 6

695278824. 8

695278825. 12

Question Number : 16 Question Id : 695278241 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $y = \tan^{-1}\left(\frac{3\cos x - 4\sin x}{4\cos x + 3\sin x}\right) + 2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{3}}{2}$ is equal to:

Options :

695278826. 3

695278827. -1

695278828. 1

695278829. 2

Question Number : 17 Question Id : 695278242 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let f be a real polynomial of degree n such that $f(x) = f'(x)f''(x)$, for all $x \in \mathbb{R}$. If $f(0) = 0$, then $36\left(f'(2) + f''(2) + \int_0^2 f(x)dx\right)$ is equal to:

Options :

695278830. 42

695278831. 46

695278832. 56

695278833. 66

Question Number : 18 Question Id : 695278243 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The area of the region $\{(x, y): y \le \pi - |x|, y \le |x \sin x|, y \ge 0\}$ is:

Options :

695278834. $1+\frac{\pi^2}{8}$

695278835. $2+\frac{\pi^2}{4}$

695278836. $\frac{\pi^2}{8}-1$

695278837. $4+\frac{\pi^2}{2}$

Question Number : 19 Question Id : 695278244 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\int_{-2}^{2}\left(|\sin x|+[x\sin x]\right)dx = 2(3-\cos 2)+\beta$, where $[\cdot]$ is the greatest integer function. Then $\beta \sin\left(\frac{\beta}{2}\right)$ equals:

Options :

695278838. 1

695278839. 2

695278840. 4

695278841. 8

Question Number : 20 Question Id : 695278245 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \left(1+x+x^2\right)\left(1-y+y^2\right), y(0) = \frac{1}{2}$. Then $(2y(1) - 1)$ is equal to

Options :

695278842. $\sqrt{3}\tan\left(\frac{11\sqrt{3}}{6}\right)$

695278843. $\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{12}\right)$

695278844. $\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$

695278845. $\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{6}\right)$

Mathematics Section B

Section Id :	69527820
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527820
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Id : 695278246 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A coin is tossed 8 times. If the probability that exactly 4 heads appear in the first six tosses and exactly 3 heads appear in the last five tosses is p, then $96p$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 22 Question Id : 695278247 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the parabola P : $y^2 = 4kx$ and the ellipse E : $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Let the line segment joining the points of intersection of P and E, be their latus rectums. If the eccentricity of E is e, then $e^2+2\sqrt{2}$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 23 Question Id : 695278248 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If $A=\frac{\sin 3^\circ}{\cos 9^\circ}+\frac{\sin 9^\circ}{\cos 27^\circ}+\frac{\sin 27^\circ}{\cos 81^\circ}$ and $B = \tan 81^\circ - \tan 3^\circ$, then $\frac{B}{A}$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 24 Question Id : 695278249 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $\overrightarrow{a_k}=\left(\tan\theta_k\right)\hat{i}+\hat{j}$ and $\overrightarrow{b_k}=\hat{i}-\left(\cot\theta_k\right)\hat{j}$, where $\theta_k=\frac{2^{k-1}\pi}{2^n+1}$, for some $n\in\mathbb{N}, n>5$. Then the value of $\frac{\sum_{k=1}^{n}\left|\overrightarrow{a_k}\right|^2}{\sum_{k=1}^{n}\left|\overrightarrow{b_k}\right|^2}$ is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 25 Question Id : 695278250 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript